

Typed Predictive-Law Operators with Residual Firewalls for Systematic Mechanism Composition

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Abstract

Continual agents should reuse and compose known mechanisms without treating every predictive mismatch as evidence for a new one. We study this problem in a controlled partially observed dynamics generator with stochastic branching, probe-conditioned aliasing, deterministic parameter refinement, and noise recalibration. An earlier additive neural head fails systematic composition: it selects withheld joint mechanisms in only 9.90% of tasks. We instead learn four shared conditional-distribution operators from anonymous single-factor groups, select their analytical type on disjoint validation backgrounds, and compose complete predictive laws. A first excluded-seed prototype obtains 99.74% selection on two withheld combinations but exposes two model-criticism failures: 13.28% false veto of valid laws and only 9.72% detection of a nonlinear residual. We derive law-conditional variances for real and imaginary characteristic residuals, studentize each moment, and add a fixed multiscale Fourier state dictionary. On five fresh seeds this reduces valid false veto from 13.07% to 5.00% and raises nonlinear detection from 33.63% to 99.75%, but misses a predeclared three-mode threshold. Separately calibrating marginal, state-coupled, and intervention-coupled channels repairs that failure on seven additional fresh seeds. The final supplied-base prototype passes all 20 fixed checks: 99.89% withheld composition, 5.51% valid false veto, 76.79–100% rejection of four incompatible families, and 73.84–100% intended route coverage. The base state, mechanism groups, operator grammar, and route semantics remain supplied. The result is therefore a reproducible component study, not open-world mechanism discovery. **Keywords:** compositional world models, conditional distributions, characteristic functions, model criticism, continual learning

1 Introduction

A change in observations does not identify a change in mechanism. Prediction can fail because a known mechanism returned, two known mechanisms co-occurred, a parameter drifted within tolerance, uncertainty changed, a hidden state was omitted, observations were corrupted, or the current model family is inadequate. A continual agent that maps all such failures to model growth will

duplicate knowledge and entangle recoverable causes.

Modular world models aim to isolate reusable mechanisms [3, 4]. Their success, however, depends on whether the model’s algebra matches the data-generating composition rule. Our preceding comparative predictive-closure experiment used factorized labels and an additive neural mixture head. It recognized many seen tuples but failed almost completely on two compatible withheld combinations. The failure motivates a narrower question:

Can single-factor predictive-law operators be learned once, composed into unseen joint laws, and audited by residual channels that identify which reversible investigation is warranted?

We make four bounded contributions.

1. We formulate four mechanisms as typed conditional-distribution operators and learn one parameter set per anonymous single-factor group across varying backgrounds.
2. We derive an analytic, law-conditional studentization for complex characteristic residual moments.
3. We partition the residuals into separately calibrated marginal, state-coupled, and intervention-coupled channels.
4. We report the complete falsification chain: an additive failure, an operator-transfer near-success, a studentization near-success, and a 20-check fresh excluded evaluation.

Characteristic functions, conditional-moment tests, heteroskedastic studentization, and multiple-testing control are established statistical tools [1, 5]. The contribution tested here is their integration with learned reusable predictive-law operators and cause-typed model criticism.

2 Typed predictive-law algebra

2.1 Conditional characteristic representation

Let $R = Y - \mu_0(X)$ be a residual around a supplied base conditional mean. For query $X = x$, intervention $A = a$, and frequency ω , define

$$\phi(\omega \mid x, a) = \mathbb{E}[\exp(i\omega R) \mid X = x, A = a]. \quad (1)$$

If conditionally independent increments Z_j add to the residual, their joint law composes by multiplication:

$$\phi_S(\omega | x, a) = \phi_0(\omega | x, a) \prod_{j \in S} \psi_j(\omega | x, a). \quad (2)$$

This is ordinary convolution in the Fourier domain, not a new probability operation.

The controlled grammar contains four operators. A deterministic refinement with slope δ has

$$\psi_{\text{ref}}(\omega | x) = \exp(i\omega\delta x). \quad (3)$$

Gaussian recalibration to scale σ has

$$\psi_{\text{cal}}(\omega) = \exp(-\sigma^2\omega^2/2). \quad (4)$$

A symmetric branch of amplitude c has

$$\psi_{\text{br}}(\omega) = \cos(\omega c). \quad (5)$$

For sign probe $a \in \{-1, +1\}$ and reliability ρ , aliasing has

$$\psi_{\text{al}}(\omega | a) = \rho \exp(i\omega ca) + (1 - \rho) \exp(-i\omega ca). \quad (6)$$

Proposition 1 (Composition error). *For true factors $\psi_0, \dots, \psi_k$ and estimates $\hat{\psi}_0, \dots, \hat{\psi}_k$, all bounded in complex magnitude by one,*

$$\left| \prod_{j=0}^k \hat{\psi}_j - \prod_{j=0}^k \psi_j \right| \leq \sum_{j=0}^k |\hat{\psi}_j - \psi_j|. \quad (7)$$

Proof. Expand the product difference as a telescoping sum. Every product multiplying one factor difference has magnitude at most one. $\square$

The bound is useful only when the factorization is valid. An interaction can violate Eq. 2 even when every factor is accurately estimated.

2.2 Anonymous operator identification

The learner receives four group identifiers but not their semantic types. For each group it fits five candidates: no operator, branch, alias, refinement, and recalibration. Parameters are shared across all training backgrounds. Candidate type is selected by conditional negative log-likelihood on disjoint validation backgrounds. The selected operators form one library; they are not refit for evaluation tasks.

Discovery observations select the lowest-NLL composition from the finite closure. Confirmation outcomes are withheld until after the library and task-level composition are frozen. A final query outcome is used only for consequence NLL.

3 Law-conditional residual firewall

3.1 Studentized characteristic moments

For predictable real basis g_t , selected law $\phi_t(\omega)$, and confirmation residual R_t , define

$$\hat{m}_{\omega,g} = \frac{1}{n} \sum_{t=1}^n g_t \{ \exp(i\omega R_t) - \phi_t(\omega) \}. \quad (8)$$

The raw prototype scaled these moments by a pooled median absolute deviation. This penalized valid high-variance laws. The selected law instead determines the conditional sampling variance. Since $\cos^2 z = (1 + \cos 2z)/2$ and $\sin^2 z = (1 - \cos 2z)/2$,

$$v_t^{\Re}(\omega) = \frac{1 + \Re\phi_t(2\omega)}{2} - \Re\phi_t(\omega)^2, \quad (9)$$

$$v_t^{\Im}(\omega) = \frac{1 - \Re\phi_t(2\omega)}{2} - \Im\phi_t(\omega)^2. \quad (10)$$

Conditional on the query design and independent confirmation draws,

$$s_{\omega,g}^{\Re} = \left\{ \frac{1}{n^2} \sum_t g_t^2 v_t^{\Re}(\omega) \right\}^{1/2}, \quad (11)$$

with the analogous imaginary expression. We divide the corresponding component of Eq. 8 by Eq. 11, using a predeclared numerical floor.

3.2 Witness dictionary and typed channels

The outcome frequencies are $\{0.5, 1, 2, 3.5, 5\}$. The state dictionary contains $1, x, x^2 - x^2$, active probe, x times active probe, and centered $\sin(kx), \cos(kx)$ for integer $k = 1, \dots, 8$. Real and imaginary components yield 210 coordinates.

A global maximum makes a ten-coordinate distribution-shape signal pay the multiplicity cost of all coordinates. We instead fix three disjoint channels:

Channel	Coordinates	Error allocation
Marginal	10	0.02
State-coupled	180	0.02
Intervention-coupled	20	0.01

Each channel receives its higher empirical quantile on a known-only calibration split. Their union is the veto. The crossing channel maps to a reversible investigation: distribution/observation audit, state-model audit, or intervention/interaction audit. It does not authorize a persistent module.

Kernel conditional-moment and conditional Stein tests provide richer, distribution-complete alternatives under their assumptions [2, 5]. Sequential kernel testing can provide anytime-valid extensions [6]. Our finite dictionary and empirical thresholds are deliberately simpler and remain fixed-horizon.

4 Benchmark

4.1 Mechanism families

The scalar state x is uniform on $[-1, 1]$. Each task has a random linear base mean and Gaussian base noise. The declared library contains base, branch, alias, refinement, recalibration, alias plus refinement, and branch plus recalibration. Two compatible combinations are withheld from operator training and calibration: branch plus refinement and alias plus recalibration.

Four ineligible families test distinct failure causes:

Table 1: Successive excluded evaluations. “Valid veto” is the fraction of withheld compatible tasks rejected by the firewall.

Stage	Sel.	Veto	Weak case
Additive head	9.90	–	composition
Shared operators	99.74	13.28	nonlinear (9.72)
Global student.	99.90	5.00	three-mode (57.13)
Typed firewall	99.89	5.51	three-mode (76.79)

1. a random-phase nonlinear residual $b \sin(\nu x + \varphi)$, with $\nu \in \{3, 5, 7\}$;
2. a three-mode successor distribution;
3. sparse signed observation corruption; and
4. a probe-associated successor whose mean interacts with x .

The word “ineligible” means outside the declared compositional closure. It does not mean a semantically new real-world mechanism.

4.2 Data separation and controls

Every seed has disjoint operator training, operator validation, known-only firewall calibration, seen evaluation, withheld evaluation, and ineligible evaluation splits. Confirmation outcomes are observed only after task selection. Query outcomes are used only for consequence evaluation.

Controls are an oracle-parameter typed library, a library restricted to seen tuples, the original pooled raw moments, and a global studentized maximum. The final protocol uses seven fresh seeds and 20 conjunctive checks fixed before execution. Exact replay rebuilds every library and dataset.

5 Results

5.1 Falsification chain

The first shared-operator prototype identifies all 12 anonymous operator types across three seeds. Mean absolute parameter errors are below 0.0041. Withheld selection is 99.74%, compared with 99.87% for oracle parameters and 0% for the seen-only control. The learned-oracle true-tuple NLL gap is 0.000059 nats. The full result nevertheless fails because one pooled firewall over-vetoes valid alias-plus-recalibration tasks and misses nonlinear residuals.

On five fresh seeds, law-conditional studentization reduces raw withheld false veto from 13.07% to 5.00%. The Fourier dictionary raises nonlinear veto from 33.63% to 99.75%. The global 210-coordinate maximum detects only 57.13% of three-mode tasks, below a fixed 60% floor. A postfailure diagnostic fixes the channel partition and allocation; it is then evaluated unchanged on seven new seeds.

5.2 Final fresh excluded result

Learned true-tuple query NLL is 0.570365, compared with 0.570270 for oracle parameters. The 0.000094 gap shows that selection accuracy is accompanied by nearly

Table 2: U4R2 result across seven fresh seeds (percent).

Metric	Mean
Seen selection	99.9575
Withheld selection	99.8884
Oracle withheld selection	99.8512
Typed calibration rejection	4.3793
Typed seen false veto	5.1871
Typed withheld false veto	5.5060
Raw withheld false veto	15.5506
No-composition withheld selection	0.0000

Table 3: Ineligible-law veto and intended route coverage (percent).

Family	Veto	Route
Nonlinear residual	100.0000	100.0000
Three-mode successor	76.7857	73.8393
Sparse corruption	100.0000	100.0000
Entangled successor	100.0000	99.9107

oracle predictive consequences. The learned library’s slightly higher selection rate than the oracle is finite-sample variation.

The three-mode veto improves 20.45 points over the global studentized control. Every one of the 20 fixed checks passes. A detached audit exactly replays all seven libraries, dataset hashes, scientific payloads, aggregate metrics, and checks.

6 What the result does and does not show

Supported component. Stable typed operators can be learned from grouped single-factor data and composed without per-task refitting in the declared generator. Conditional studentization transfers calibration across valid laws with different predictive variances. Separate residual channels retain power against different incompatibilities and expose an investigation type.

Supplied structure. The base mean, scalar state, mechanism group boundaries, candidate operator types, compatibility and ordering grammar, witness dictionary, channel partition, and action semantics are provided. Mechanism parameters are stationary. Consequently the experiment does not solve representation learning or open-world ontology construction.

Statistical boundary. The empirical quantiles are not an exact finite-sample family-wise theorem. Task selection is disjoint from confirmation, but the test is fixed-horizon. A continual system needs predictable witnesses, repeated-test accounting, tolerance-aware approximate nulls, and evidence tied to the candidate later admitted.

Domain boundary. The benchmark is synthetic, low-dimensional, and CPU-small. There is no pixel perception, language, recognized control suite, natural distribution shift, hardware measurement, or comparison with large foundation models.

7 Related work

Independent-mechanism world models learn modular dynamics and recombine them across environments [3, 4]. This study differs by operating on complete conditional laws and by making model criticism an explicit typed gate, but its supplied grammar is substantially narrower.

Conditional moment restrictions can be tested through maximum moments in an RKHS [5]; conditional density models can be criticized with kernel Stein operators [2]; and conditional characteristic functions have long supported specification tests for stochastic processes [1]. Sequential kernel goodness-of-fit uses test-by-betting for streaming validity [6]. These works delimit our contribution: neither the spectral residual nor the test machinery is new. The open research question is whether they can govern learned modular world-model repair under adaptive, open-world operation.

8 Next experiments

The next stage replaces the supplied base mean with a discovery-trained nonlinear world model. Its uncertainty must enter Eq. 11; otherwise base-model error will be misclassified as operator incompatibility. The evaluation should include an oracle-base control, a deliberately mis-specified base, hidden-state aliasing that requires an active probe, parameter drift, and recurrent regimes.

Subsequent work must replace grouped operator identities with discovery-frozen proposals, scale composition retrieval beyond a nine-law closure, and use sequential evidence budgets before persistent library writes. Only then is a recognized embodied domain informative.

9 Reproducibility

The final source commit is a5ea967b9627aa6c3cd46812d01bddde2136d6e5. The result artifact SHA-256 is 81F95D78A56A916644B59A4510B6AD6F475B39442FD854E19F99D25B13DDB006; the runtime-independent science hash is 40ED5B513AE507B01A82063AD8C070330FC04E55612A235620B9346186BD5AD5. The detached replay audit passes 10/10 checks. Protocols, configs, code, tests, results, audits, negative predecessors, and claim boundaries are stored as separate records.

10 Conclusion

Systematic composition requires the learning architecture to match the composition law, and safe continual reuse requires criticism of the complete composed distribution. In the controlled benchmark, learned conditional-distribution operators and a typed residual firewall satisfy both requirements on fresh excluded seeds. The strongest remaining limitation is also the central open problem:

the representation, base model, ontology, and repair semantics are still supplied. The next useful experiment is therefore not a larger version of the same generator, but a learned nonlinear world model in which representation error competes honestly with operator reuse, refinement, recalibration, and interaction.

Research-assistance disclosure. An AI coding and writing assistant supported code, experiments, mathematical organization, literature search, and drafting. The human author is responsible for independently checking every derivation, citation, result, artifact, license, disclosure, and final submission.

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